

Abigail Paulus

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Ambitious, detail-oriented mechanical engineering student focused on drafting, aspiring to be a design engineer

Education

B.S. in Mechanical Engineering, Minor in German | Florida Atlantic University, Boca Raton, FL May 2028

Awards: \$1,000 CSYS Scholarship (2024), Outstanding Engineering Achievement Award (Fall 2024 & Fall 2023), Cambridge AICE Diploma (2023)

Skills & Qualifications

Certifications: Solidworks, DraftSight, AutoCAD, Inventor, Fusion360, Excel, Word, PowerPoint, Photoshop, HTML, CSS, JS, Test Out IT

Technical: Soldering, FDM 3D Printing, Calipers, Power Tools, Drill Press, Woodworking, Weldments, CNC Programing, GD&T

Software: SketchUp, RapidHarness, Siemens NX, Solidworks PDM, ERP, AS400, Illustrator, Inkscape, Cura Slicer, Altium Designer, Blender

Professional Experience

Supply Chain Documentation Intern | Dometic, Pompano Beach, FL Oct 2025 – Present

- Organize import, export, and finished goods documentation of 13,200+ exported products and components in Excel
- Maintaining documentation in Word and Adobe Acrobat for audit trail in efforts to recover est. \$500k in tariff reimbursement

Programing Teacher (Contract) | Code/Art, Davie, FL Feb 2024 – Present

- Instructing 10+ students within 1–2-hour lesson using programs such as: Scratch, p5.js, Tinkercad, JavaScript, HTML/CSS
- Verifying comprehension of lessons, facilitating attendance, and establishing a supportive space for students to thrive

Product Design Engineer Intern | Dometic, Pompano Beach, FL May 2025 – Sept 2025

- Prepare, maintain, and update 75+ drawings for components, layouts, assemblies in proper sequence and order
- Administering, maintaining, and updating DCR, PSR, RCR, NRP, ECN, ERC, and other product information
- Tracking and assigning part numbers and BOM for products up to date to maintain product information documentation
- Communicating, through Outlook, maintaining communication and clarity regarding NPR drawings or updates

Mechanical Engineer Intern | Q'Straint, Oakland Park, FL June 2024 – Aug 2024

- Designed metal support brackets for in-house production using SolidWorks' sheet metal tool and design intent
- Tracked 200+ unused inventory parts, finalized 10+ drawings, fulfilled 2 work orders
- Drafted 25-page service manual, increasing technicians' efficiency, reduce costs, improving convenience
- Enhanced UX/UI, optimized, hosted on GitHub (plans to use SQL and PHP, adding features to improve communication)

CAD Engineer Intern | Q'Straint, Oakland Park, FL June 2023 – July 2023

- Updated 10+ outdated technical drawings to match current standards, replacing to use existing inventory, reducing cost
- Authored 6 pages of instruction manual using Word, also serving as template for similar components across product line
- Use up existing inventory by developing McMaster Carr replica, reducing cost, and efficiently filtering 150+ small hardware parts
- Edited/enhanced 16 pages of introductory engineering handbook in OneNote, facilitating smoother onboarding process

Projects

Designer | Electric Campervan Conversion Oct 2025 – Present

Developing modular plumbing, electrical, and HVAC systems for electric Chevy Brightdrop 600 van chassis to convert into campervan, designed using Siemens NX, with 15kW of additional onboard power, 35 gallons of water, charged by 600W of solar

Product Developer | Digital Audio Player July 2024 – Present

MP3 player reverse engineered using Raspberry Pi with custom PCBA and CNC machined chassis practicing DFM/DFA (design for manufacturing/assembly), analyzing consumer needs, prioritizing features most valuable and reiterating different prototypes

Team Lead | Blind Spot Sensor for Motorcycle/Bicycle Dec 2023 – Feb 2024

Designed concept in Fusion360, with 4 FDM prototypes, ensuring proper fit of existing components found in premade starter kit to develop Arduino based modular system that detect objects within a biker's blind-spot, with vibrating motor module on the helmet

Project Manager | Bar-top Display Stand for Teacher's Lounge Mar 2023 – Oct 2023

Arranged cut list using Excel and drafted assembly drawings within SketchUp, while managing team of 3 with effective communication, to design and build custom wooden bar-top, optimizing material selection, leaving 30% of \$1000 budget

Leadership Experience

Technical Outreach and Community Help (TORCH) Chair | National Society of Black Engineers (NSBE) June 2025 – Present

Internal Development Chair/Secretary | American Society of Mechanical Engineers (ASME) Jan 2025 – Present

Academic, Professional & Athletic Chair | Alpha Omega Epsilon (AOE) Jan 2025 – Present

Volunteer Coach (Seasonal) | Coral Springs Youth Soccer (CSYS) Nov 2021 – Present

President/Vice President | Southeastern Consortium for Minorities in Engineering (SECME) June 2022 – 2024

Active Club Membership: SWE, SHPE, Women in Engineering and Computer Science, Technology & Aerospace Club (SEDS)